

## CHAPTER 2

### LITERATURE SURVEY

- **Acharya, U. R., Oh, S. L., Hagiwara, Y., Tan, J. H., & Adeli, H. (2018). Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals.[2]**

An encephalogram (EEG) is a commonly used ancillary test to aid in the diagnosis of epilepsy. The EEG signal contains information about the electrical activity of the brain. Traditionally, neurologists employ direct visual inspection to identify epileptiform abnormalities. This technique can be time-consuming, limited by technical artifact, provides variable results secondary to reader expertise level, and is limited in identifying abnormalities. Therefore, it is essential to develop a computer-aided diagnosis (CAD) system to automatically distinguish the class of these EEG signals using machine learning techniques. This is the first study to employ the convolutional neural network (CNN) for analysis of EEG signals. In this work, a 13-layer deep convolutional neural network (CNN) algorithm is implemented to detect normal, preictal, and seizure classes. The proposed technique achieved an accuracy, specificity, and sensitivity of 88.67%, 90.00% and 95.00%, respectively.[2]

#### **Advantages:**

- Introduces a deep convolutional neural network (CNN) for automated seizure detection using EEG signals, reducing the need for manual interpretation.
- Demonstrates high accuracy in detecting seizures, showcasing the potential of deep learning in medical diagnostics.

#### **Disadvantages:**

- The model's performance may vary with different EEG datasets, indicating potential challenges in generalizability.
- Deep learning models require substantial computational resources and large labelled datasets for training.

- **Rathi, S., Patel, V., & Agrawal, R. (2023). [4] Deep learning and machine learning in psychiatry: A survey of current progress in depression detection, diagnosis, and treatment.**

Recent advances in brain and mental health research are driven by technologies like machine learning, deep learning, and AI, enabling more personalized and precise approaches in diagnosing and treating depression. [4] Precision psychiatry uses these tools to tailor care, identify at-risk individuals, and challenge traditional diagnostic categories through unsupervised learning.[4] While AI supports all treatment phases and offers evidence-based alternatives, current research lacks empirical proof of better patient outcomes. To advance clinical use, there's a need for rigorous validation, better data access, interdisciplinary collaboration, and standardized definitions.

#### **Advantages:**

- Explores the application of deep learning and machine learning in detecting and treating depression, emphasizing precision psychiatry.
- Discusses the potential of AI to provide personalized mental health care.

#### **Disadvantages:**

- Notes that many AI models lack empirical validation in clinical settings.
- Highlights challenges like data heterogeneity and the need for standardized definitions in psychiatric research.

- **Saha, S., & Dasgupta, R. (2022). Hybrid CNN-LSTM architecture for EEG-based human activity recognition.[5] In Proceedings of the International Conference on Intelligent Computing and Smart Communication 2022**

Human Activity Recognition (HAR) involves predicting a person's activity using sensor data from wearable devices. We use a hybrid CNN-LSTM model to improve recognition accuracy, especially during activity transitions. [5] CNN handles spatial features from raw data, while LSTM captures temporal transitions. The model is trained on HAPT and HAR datasets, with performance optimized through hyper parameter tuning. Evaluation metrics like precision, recall, and F1-score show improved accuracy with optimized settings.

**Advantages:**

- Combines CNN and LSTM architectures to capture both spatial and temporal features in EEG data, enhancing human activity recognition.
- Demonstrates improved accuracy over traditional methods.

**Disadvantages:**

- Hybrid models can be computationally intensive, requiring more resources for training and deployment.
  - May face challenges in real-time applications due to increased complexity.
- **Chen, S., Yu, Y., & Pan, J. (2023). [6] MADNet: EEG-Based Depression Detection Using a Deep Convolution Neural Network Framework with Multi-dimensional Attention. In Artificial Neural Networks and Machine Learning – ICANN 2023[6]**

Major depressive disorder (MDD) is a common and serious mental health problem that has received increasing attention from both researchers and clinicians. Electroencephalography (EEG)-based automatic diagnosis of MDD has been explored in previous studies, but feature extraction remains an area for improvement. We propose a novel deep learning approach for the automated detection of MDD. For feature extraction, a multi-dimensional attention mechanism is used to extract features from a simple pre-processed EEG signal to enhance spatio-temporal feature learning.[6] We randomly divided EEG data from 26 MDD patients and 26 healthy controls into training and testing sets for network training and testing, respectively. Our method achieved the highest accuracy of 95.14% in subject-based data classification experiments compared to other methods. As a result of its simplicity and convenience, the method proposed in this study has the potential to be utilized as a tool for assisting in the daily diagnosis of depression.

**Advantages:**

- Introduces MADNet, a deep CNN framework with multi-dimensional attention mechanisms for EEG-based depression detection.
- Achieves high accuracy, showcasing the effectiveness of attention mechanisms in enhancing model performance.

**Disadvantages:**

- The complexity of the model may hinder its interpretability and real-time application.
- Requires extensive computational resources for training and deployment.

- **Ahmadlou, M., Adeli, H., & Adeli, A. (2012). [7]Fractality analysis of frontal brain in major depressive disorder. *International Journal of Psychophysiology***

While emotional stimuli are known to affect pain perception, this study explored how pain influences emotion processing. Involving 30 participants, researchers measured muscle responses to happy and angry faces during painful and non-painful stimulation. Pain slowed both smiling and frowning responses, particularly reducing incorrect frowning responses to happy faces, but had no effect on reactions to angry faces or emotional ratings.[7] These findings suggest pain selectively alters the processing of happy faces, potentially impacting social interactions during pain.

**Advantages:**

- Utilizes fractal analysis to study EEG signals in patients with major depressive disorder, offering insights into brain dynamics.
- Provides a novel perspective on EEG signal analysis beyond traditional linear methods.

**Disadvantages:**

- Fractal analysis techniques may be sensitive to noise and require careful preprocessing.
- The study's findings may need further validation across diverse populations.

- **Acharya, U. R., Sudarshan, V. K., Adeli, H., Santhosh, J., Koh, J. E., & Adeli, A. (2015).[10] A novel depression diagnosis index using nonlinear features in EEG signals. *European Neurology*.**

Sex chromosome aneuploidies range in incidence from rather common to exceedingly rare and have a variable phenotype. We report 2 patients with sex chromosome aneuploidies who developed severe aplastic anemia requiring treatment. The first patient

had tetrasomy X and presented at 9 years of age, and the second patient had trisomy and presented at 5 years of age. Although aplastic anemia has been associated with other chromosomal abnormalities, sex chromosome abnormalities have not been traditionally considered a risk factor for this condition. A review of the literature reveals that at least one other patient with a sex chromosome aneuploidy (45,X) has suffered from aplastic anemia and that other autosomal chromosomal anomalies have been described.[10] Despite the uncommon nature of each condition, it is possible that the apparent association is coincidental. A better understanding of the genetic causes of aplastic anemia remains important.

**Advantages:**

- Introduces a novel depression diagnosis index using nonlinear features in EEG signals, enhancing diagnostic accuracy.
- Combines multiple nonlinear metrics to capture complex brain dynamics associated with depression.

**Disadvantages:**

- The approach may require extensive computational resources for feature extraction and analysis.
  - Clinical applicability needs further exploration through larger studies.
- **Alhagry, S., Fahmy, A. A., & El-Khoribi, R. A. (2017).[17] Emotion recognition based on EEG using LSTM recurrent neural network. International Journal of Advanced Computer Science and Applications.**

Emotion is the most important component in daily interaction between people. Nowadays, it is important to make the computers understand user's emotion who interacts with it in human-computer interaction (HCI) systems. Electroencephalogram (EEG) signals are the main source of emotion in our body.[17] Recently, emotion recognition based on EEG signals have attracted many researchers and many methods were reported. Different types of features were extracted from EEG signals then different types of classifiers were applied to these features. In this paper, a deep learning method is proposed to recognize emotion from raw EEG signals. Long-Short Term Memory

(LSTM) is used to learn features from EEG signals then the dense layer classifies these features into low/high arousal, valence, and liking. DEAP dataset is used to verify this method which gives an average accuracy of 85.65%, 85.45%, and 87.99% with arousal, valence, and liking classes, respectively. The proposed method introduced high average accuracy in comparison with the traditional techniques.

**Advantages:**

- Utilizes LSTM to capture temporal dependencies in EEG signals effectively.
- Demonstrates promising performance in emotion classification using deep learning.

**Disadvantages:**

- Relatively small dataset, which may affect generalizability.
- Focused only on emotion detection, not suitable for broader psychiatric diagnosis.

- **Y. Ding, X. Chen, Q. Fu, S. Zhong [9], A Depression Recognition Method for College Students Using Deep Integrated Support Vector Algorithm.” IEEE Access (2020).**

A deep integrated Support Vector Machine (SVM) framework was developed to identify depression levels among college students using EEG signals. The hybrid model combines deep feature extraction with SVM classification to enhance recognition accuracy. Deep learning layers automatically extract high-level EEG features, while the SVM classifier provides precise decision boundaries, leading to improved generalization even with limited data. Experimental evaluation showed that the integrated method achieved better accuracy and robustness than traditional SVM and standalone deep networks. The work emphasizes the potential of combining deep learning and classical machine-learning approaches for effective EEG-based mental-health screening.

**Advantages:**

- Integration of deep learning and SVM improves classification performance.
- Efficient handling of smaller datasets through enhanced feature extraction.

- Suitable for practical mental-health monitoring among college students.

**Disadvantages:**

- High computational cost due to hybrid network design.
  - Sensitive to parameter selection and tuning.
  - Generalization limited to the specific demographic studied.
- **M. Bachmann, L. P"oeske, K. Kalev, K. Aarma, A. Lehtmets, P. " O"opik, J. Lass, H. Hinrikus,[10] Methods for classifying depression in single channel EEG using linear and nonlinear signal analysis, Comput. Methods Programs Biomed. 155 (2018)**

Linear and nonlinear signal-analysis techniques were employed to classify depressive and normal conditions from single-channel EEG recordings. The approach extracted both spectral and nonlinear parameters such as power spectral density, fractal dimension, and entropy to capture differences in brain dynamics associated with depression. Support Vector Machine (SVM) classifiers were then applied to the extracted features to separate the two groups. The results demonstrated that even single-channel EEG can provide valuable diagnostic information when analyzed using combined linear and nonlinear measures. This work highlights the potential for developing portable and simplified EEG-based systems for mental-health screening.

**Advantages:**

- Effective combination of linear and nonlinear EEG features for accurate classification.
- Demonstrates feasibility of depression detection using minimal EEG channels.
- Reduces complexity and cost of data acquisition.

**Disadvantages:**

- Limited spatial information due to single-channel recording.
- Accuracy slightly lower than multi-channel or deep-learning models.
- Small dataset may restrict generalization of results.

- **S.D. Puthankattil, P.K. Joseph, Classification of EEG signals in normal and depression conditions by ANN using RWE and signal entropy, J. Mech. Med. Biol. 12 (2012) 1240019[9].**

An Artificial Neural Network (ANN) model was utilized to classify EEG signals recorded from normal and depressive subjects using Relative Wavelet Energy (RWE) and signal entropy as distinguishing features. The extracted RWE values captured frequency-domain energy variations across different EEG sub-bands, while entropy represented the randomness and complexity of neural activity[9]. By training the ANN on these combined features, the model effectively differentiated depressive EEG patterns from normal ones. The approach provided an early example of machine learning applied to mental-health signal processing and demonstrated that wavelet-based energy and entropy measures are valuable indicators of depression-related changes in brain activity.

#### **Advantages:**

- Combines wavelet energy and entropy features for enhanced diagnostic accuracy.
- Simple ANN model offers fast training and easy implementation.
- Establishes the foundation for later hybrid EEG-based depression detection systems.

#### **Disadvantages:**

- Lacks temporal sequence modeling of EEG signals.
  - Performance depends on manual feature extraction and preprocessing.
  - Accuracy limited compared to modern deep-learning approaches.
- **O. Faust, Y. Hagiwara, T.J. Hong, O.S. Lih, U.R. Acharya, Deep learning for healthcare applications based on physiological signals: A review, Comput. Methods Programs Biomed. 161 (2018) 1–13[8].**

A comprehensive review of deep-learning applications in healthcare was presented, focusing on physiological signals such as EEG, ECG, and EMG. The paper summarized the major architectures—Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and hybrid frameworks—used for analyzing medical time-series data.

Special attention was given to the role of deep learning in automating feature extraction, improving diagnostic accuracy, and enabling real-time disease detection. The review highlighted how deep models surpass traditional machine-learning techniques in handling complex biomedical patterns, while also addressing challenges such as data scarcity, model interpretability, and computational cost[8].

**Advantages:**

- Provides a broad overview of deep-learning models applicable to healthcare signal analysis.
- Demonstrates the ability of neural networks to automatically learn discriminative physiological features.
- Establishes the foundation for modern deep-EEG depression-detection frameworks.

**Disadvantages:**

- Primarily conceptual; lacks detailed quantitative evaluation.
- High computational requirements limit deployment in low-resource settings.
- Challenges related to data privacy, interpretability, and clinical validation remain unresolved.

# **CHAPTER 3**

## **SYSTEM ANALYSIS**